

CSAI 810 Section 001 – Project Definition

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Automated Multimodal Agent: PDF to Narrated PowerPoint

Short Description of Goals and Limitations

The primary goal of this project is to build an advanced, agent-based system that autonomously converts a PDF document (such as a lecture or book chapter) into a fully narrated PowerPoint presentation. Unlike a standard linear pipeline, this system will employ an intelligent agent workflow to parse content, plan slide structures, generate summaries, and synthesize voice narration using Hugging Face models.

Limitations: The project addresses the technical limitation of the python-pptx library, which lacks a direct API for embedding audio, by implementing a workaround that converts audio to MP4. The system is constrained to a 30-hour development cycle and will focus on functional integration rather than high-end visual aesthetics (e.g. animations).

What will we achieve (learn + produce) and what is your prior knowledge?

Produce: We will produce a functional Python application consisting of multiple specialized agents (Parser, Planner, Generator, TTS) and a final .pptx file with embedded, auto-playing narration for each slide.

Learn: This project serves as a practical deep-dive into the **Generative AI Agentic Workflow** (Think → Plan → Act), moving beyond simple prompt engineering. We will learn to integrate multimodal systems (Text-to-Speech) and manage complex file manipulations programmatically.

Prior Knowledge: We possess a foundational knowledge of Python and standard machine learning pipelines. However, this project challenges us to implement autonomous decision-making logic and multimodal integration, areas where we have limited prior practical experience.

Why do we want to do this project?

We chose this project because it moves beyond traditional "demo" applications to create a system that exhibits genuine intelligent behavior. Academically, it presents a unique challenge in coordinating multiple AI models (LayoutLM, Mistral/Flan-T5, Bark/Sunno) to solve a complex task. Practically, it offers high "Human Value" by drastically reducing the time educators spend on lecture preparation and enhancing accessibility for students with hearing impairments through synchronized text and audio.

Final Deliverable and Target Users

Deliverables:

- **lecture.pptx:** A complete, self-contained presentation file where every slide includes auto-playing audio narration (embedded as an invisible/audio-only MP4).
- **Source Code:** A Python script/Notebook containing the agent logic.
- **Documentation:** A README explaining the architecture and usage instructions.

Target Users: The primary users are **educators** and **academic institutions**, particularly those with limited resources who need to scale their content creation, as well as students requiring accessible study materials.

Project Boundaries (Feasibility within 30 Hours)

Included (In Scope):

- Single PDF input
- English TTS narration embedded as MP4
- Output of 5-10 slides
- Basic summarization and slide-planning agent

Excluded (Out of Scope):

- Video generation (avatars or lip-syncing)
- Advanced slide animations or transitions
- Multi-language support or real-time streaming
- Advanced visual design or interactive learning features